MIL-T-60156(MU)
AMENDMENT 3
20 February 1968
SUPERSEDING
AMENDMENT 2
1 July 1965

MILITARY SPECIFICATION

TELESCOPE: XM119

This amendment forms a part of Military Specification MIL-T-60156(MU), dated 12 November 1964.

* Title: Delete "XM119" and substitute "M119" and make this same change wherever "XM119" is used in this specification.

* Paragraph 3.3.2, 3.16.2, 3.16.3, 3.16.4, 3.16.4.1 and 4.5.9: Delete "-40°" and substitute "-65°".

Paragraph 3.7.1.2: Delete "circle" and substitute "square".

Paragraph 3.7.3: Delete "6 mils in depression and 9 mils in elevation" and substitute "9 mils in depression and 6 mils in elevation".

Paragraph 3.11: Add at end of paragraph: "at one established focus setting".

Paragraph 3.13: Delete "0.2 mil" and substitute "2 percent".

Paragraph 3.14: Add "Parallax is nominally zero at 1200 meters. Parallax between reticles shall not exceed 0.05 mil".

Paragraph 3.15.1: Delete paragraph and substitute:

"Missile reticle.- The horizontal reticle line of the missile reticle shall be parallel to the image of a horizontal target line within 30 minutes of arc."

Paragraph 3.15.2: Delete paragraph and substitute:

"3.15.2 Conventional reticle.- The representative 1600 meter horizontal reticle line (total length of dashes and spaces) of the conventional reticle shall be parallel to the image of a horizontal target line within 30 minutes of arc."

Paragraph 3.15.3: Delete paragraph and substitute:

"3.15.3 Variation of settings.- The angular difference between the horizontal reticle line of the missile reticle and the 1600 meter line of the conventional reticle shall not exceed 30 minutes of arc."

* Paragraph 3.16.1: Delete and substitute:

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"3.16.1 Diopter knob disengagement.- The force required to disengage the diopter knob from its normal position shall be 4 ± 2 pounds at standard ambient temperature (+60° to +90°F), and shall be 6 ± 4 pounds at -65° and +150°F."

* Add the following paragraph:

"3.16.1.1 Diopter knob rotation.- The torque required to rotate the disengaged diopter knob shall be 2 ± 1 inch pounds at standard ambient temperature (+60° to 90°F), and shall be 3 ± 2 inch pounds at -65° and +150°F."

* Paragraph 4.2.2, Table I: Delete Characteristic "120"; add the following characteristics:

"130.	Diopter knob disengagement	3.16.1	4.5.8
131.	Diopter knob rotation	3.16.1.1	4.5.8.1."

* Paragraph 4.4.2.3.3, 1st sentence: Delete "...minimum range from zero to twenty pound inches, the accuracy of which shall be plus or minus one-quarter pound inch..." and substitute "...minimum range from zero to ten inch pounds, the accuracy of which shall be plus or minus two inch ounces..."

* Add new paragraph:

"4.4.2.3.4 Special testing to test force requirements.- Testing equipment for measuring direct force (pull) with a minimum range of zero to twelve pounds, the accuracy of which shall be plus or minus four ounces throughout the entire zone of excursion, shall be utilized. The testing equipment shall be equipped with suitable adapters to accommodate testing of the diopter knob and boresight knobs."

* Paragraph 4.5.2, 4th sentence: Delete "4.5.7" and substitute "4.5.8".

Paragraph 4.5.7.1.2: Delete "circle" and substitute "square".

Paragraph 4.5.7.3: Delete "3.7.2" and substitute "3.7.3".

Paragraph 4.5.7.9, 2nd sentence: Delete "and the resolving power chart shall contain four sets of lines as shown in Figure 1 therein", and substiture "except that the focus of the auxiliary telescope shall be established at one best focus setting for simultaneous viewing of the four meridians of the resolving power chart. The resolving power chart shall contain four line sets as shown in Figure 1 of MIL-O-13830."

Paragraph 4.5.7.12: Delete last sentence and substitute "Parallax between each reticle and the target image, and parallax between reticles shall not exceed the applicable tolerance specified in 3.14.

Paragraph 4.5.7.13: Delete paragraph and substitute:

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"4.5.7.13 Parallelism of missile reticle and image.- Sight through the eyepiece of the telescope and obtain coincidence of one end of the telescope's horizontal reticle line with that of the horizontal target line. Then observe coincidence at the opposite end of the reticle line. The horizontal reticle line shall be parallel to the horizontal target line image within the tolerance specified in 3.15.1, as measured on the target. Deviation shall be noted and recorded."

Paragraph 4.5.7.14: Delete paragraph and substitute:

"4.5.7.14 Parallelism of conventional reticle and image.- Project the conventional reticle pattern into the optical system of the telescope by means of the reticle selector switch. Sight through the eyepiece of the telescope and obtain coincidence of one end of the telescope's 1600 meter horizontal reticle line (total length of dashes and spaces) with that of a horizontal target line. Then observe coincidence at the opposite end of the reticle line. The horizontal reticle line shall be parallel to the horizontal target line within the tolerance specified in 3.15.2, as measured on the target. Deviation shall be noted and recorded."

* Paragraph 4.5.8: Delete and substitute:

"4.5.8 Diopter knob disengagement.- Testing equipment conforming to the requirements of 4.4.2.3.4 shall be utilized for this test. The force (pull) required to disengage the diopter knob from its normal position shall be within the values specified in 3.16.1 for the temperatures indicated."

* Add new paragraph:

"4.5.8.1 Diopter knob rotation.- Testing equipment conforming to the requirements of 4.4.2.3.3 shall be utilized for this test. The running torque required to rotate the disengaged diopter knob throughout its range of movement shall be within the values specified in 3.16.1.1 for the temperature indicated."

* Paragraph 4.5.11: Delete and substitute:

"Boresight knob disengagement.- Testing equipment conforming to the requirements of 4.4.2.3.4 shall be utilized for this test. The force (pull) required to disengage each boresight knob from its normal position shall be within the values specified in 3.16.4 for the temperature indicated."

* Paragraph 4.5.12: Delete 2nd and 3rd sentences and substitute:

"The running torque required to rotate each disengaged boresight knob throughout its range of movement shall be within the values specified in 3.16.4.1 for the temperature indicated."

* The outside margins have been marked to indicate where additions to the previous amendment have been made. This has been done as a convenience only

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